

Semantic Afterlife: Lock Occupancy After the Prompt Leaves the Window

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Abstract

A language model whose input is truncated to the last W generated tokens continues after the original prompt has left that window. We ask what remains in the trajectory: a unique stationary law, a temperature-robust cycle, a domain-tagged lock, or diffusion. The object is the *afterlife*: the segment after seed eviction, not the prompt-conditioned transient.

On one instruct generator (`or-qwen3-8b`) under a re-prompt protocol (P1), a textual fixed-point lock is the typical $T=0.3$ outcome through twelve turnovers of $W=4096$. Last-band occupancy retains an ensemble domain gap (F4 distinguishability); that gap is not recovered prompt memory (H2). That occupancy result is a subsection, not the title claim: the third space is an NHST whisker, not architecture-independent robustness. The lock is not universal across hosted generators, is absent at $T=1.5$ for $W=4096$ on this model, and is not a validated Markov state model. We did not measure a sliding native KV cache. Degeneracy is the sample, not a filter that was applied and then forgotten.

1 Introduction

Let $X_t \in V^W$ be the last W generator tokens, $Y_t \sim P_\theta(\cdot | X_t)$, and $X_{t+1} = \text{Tail}_W(X_t \oplus Y_t)$. After roughly one window of generated tokens the seed is physically absent from the model’s input. Call that eviction the *horizon*, and the subsequent trajectory the *afterlife*. The scientific question is not whether a model can follow a prompt while the prompt is present. It is whether, after eviction, the trajectory retains seed identity, collapses onto a model-specific lock, or diffuses.

That question is easy to answer wrongly. A looping trajectory occupies one point in representation space and reports a confined mean-squared displacement for reasons that have nothing to do with semantics. This project recorded that error once and does not repeat it: degeneracy is labelled before geometry is read, and a looping row is not sold as a confined basin.

The experiments that exist are Stages 0–6 of a pre-registered plan (`docs/research-plan.md`). Stage 3 asked for metastable semantic states via VAMP/PCCA+ and did not obtain a validated MSM. Stage 4 asked for a temperature window of non-looping confined motion (H5) and did not find it. Stage 5–6 asked whether, *given* the lock, last-band occupancy still shows an ensemble domain gap and whether replica pairs collapse. The last-band result is distinguishability, not recovered prompt memory (H2). Sign agreement holds in three embedding spaces on this generator, protocol, and temperature; the third space is an NHST whisker, not architecture-independent robustness. They do not establish H1, H5, or a result for language models as a class.

Section 2 places the afterlife against verified related work. Section 3 defines P1, the cost law, degeneracy, and the two occupancy tests used later. Section 4 reports stages in the order they were run. Section 5 lists what the design cannot establish. Section 6 separates established claims from parked ones.

2 Related work

We cite only sources marked VERIFIED in the project literature log. Unverified items are not used as support.

Zekri et al. [2024] study autoregressive generation as a Markov chain on the context and, under contraction of the next-token kernel in total variation, prove existence of a unique stationary distribution together with exponential mixing. That theorem is the right null for a *unique* long-run law. It is not a measurement of whether our trajectories occupy one lock or several domain-tagged locks, and it does not replace an occupancy test after eviction.

Wang et al. [2025] report decoding cycles that remain under temperature sampling, and treat repetition as a dynamical regime rather than a decoding bug. Our Stage 2 comparison is the opposite direction on the model axis: at $T=0.3$, `or-gemma-4-31b` produced 0/8 textual fixed points while `or-gpt-oss-120b` produced 8/8 (Section 4.3). Cycle robustness to temperature in their setting is not a licence to call our lock temperature-universal; Stage 4 already falsifies that reading on `or-qwen3-8b` at $T=1.5$, $W=4096$.

Geng et al. [2026] argue that raising temperature lengthens transients before a stable regime. Stage 4 is consistent with a later onset of non-looping behaviour at $T=1.5$, but that stage did not estimate a mixing time, and the non-looping rows are subdiffusive rather than a validated stationary sample.

Ko and Geiping [2026] describe interlocutor-conditioned attractors in multi-agent dialogue. Our protocol has no interlocutor. A lock that appears without a second speaker is a different object; we do not import their attractor language as a finding about our trajectories.

Chen et al. [2026] separate horizon, context, and memory as distinct axes. We impose W by truncation (P1) and begin the scientific object after the seed has left that window. That is the intended delta relative to prompt-conditioned long-context work: the afterlife is not another name for in-window reasoning.

Markov-state and VAMP methodology from molecular dynamics [Wu and Noé, 2020, Paul et al., 2019] is the toolkit Stage 3 applied to embeddings. The toolkit is not the result. Stage 3 did not validate an MSM on these trajectories (Section 4.4).

3 Protocol

3.1 Generator, window, and P1

All occupancy numbers in Sections 4.6–4.7 use `or-qwen3-8b` (Qwen3-8B-Instruct via OpenRouter; served provider `Alibaba` on every completed occupancy step), $W=4096$, $T=0.3$, block $B=512$, analysis chunks of 1024 generator tokens, non-overlapping, and twelve turnovers of W after the seed. The continuation mechanism is `raw_completion`, not `assistant_prefill` and not `chat_instructed`. Protocol P1 is the re-prompt: the last W tokens of generated text are re-encoded and sent as a new prompt. P1 and the continuation mechanism are distinct axes (glossary). Stage 2 already measured a prefill-versus-raw arm on this generator; occupancy does not reuse that arm as its protocol. The input construction is not a sliding native KV cache. The gap is named in Section 5 and parked in ADR-0017; it is not treated as measured.

A *turnover* is T/W in generator tokens. The *horizon* is the step at which the seed has fully left the window. Last-band occupancy uses the final complete turnover band, not a mean over the whole trajectory.

3.2 Cost law

Hosted spend follows the recorded ledger, not a model card. Stage 0 capability audit cost \$0.0128. Stage 1 generation billed \$6.32 on `s1-pilot-core-20260830T134147Z-f1cce9ec`. Stage 4 hosted generation billed \$3.4383 (`s4-w4096-new-temps-20260904T103121Z-589c8eb1, s4-w8192-20260904T120057Z-ce82ce55`). Stage 5 generation billed \$1.3385 (`s5-lock-occupancy-20260905T1`). Stage 6 used cached Gemini embeddings at hosted \$0. Project hosted spend at Stage 6 close was \$16.34 of a \$200 ceiling. No Stage 7 generation or embedding was run.

3.3 Degeneracy is the sample

A trajectory is labelled a textual fixed point when calibrated repetition statistics cross a threshold derived from a human-prose reference corpus: mean 3-gram Jaccard 0.083 and late-window Jaccard 0.0122. Those two numbers are not retuned in this manuscript.

On the Stage 5–6 occupancy panel, 19/20 domain trajectories and 1/2 love trajectories meet the lock rule; 10/10 domain seeds have at least one lock. Occupancy is therefore a statement about locked trajectories, not about a clean non-looping subsample. Geometry on a looping row is not interpreted as a confined semantic basin. Love is one of the ten F4 domain seeds, not a held-out class.

3.4 Occupancy tests F4 and F6

Let D_{within} be the mean cosine distance among last-band centroids of trajectories that share a seed domain, and D_{between} the mean cosine distance among last-band centroids from different domains. The domain gap is $\Delta_{\text{dom}} = D_{\text{between}} - D_{\text{within}}$. Domains are *separated* when the bootstrap 95% CI for Δ_{dom} excludes 0 from above.

For each matched replica pair, $\Delta_{\text{pair}}(t)$ is the cosine distance between the two centroids in turnover band t , minus the same quantity in band 0. The pair is operationally *collapsed* when the last-band CI for Δ_{pair} includes 0. That operational definition is not occupancy of one shared lock: a CI that includes 0 can be an underpowered non-detection, and a last-band matrix can still show two within-seed diagonals.

F4 and F6 are assembled from occupancy sidecars (`scripts/assemble_stage6_occupancy.py`), not from a CLI `analyze separation` pass and not from a twins table with `scope=all`.

4 Results

4.1 Stage 0: the representation stack exists; hosted bases do not

Three embedding endpoints accepted the audit texts, returned the advertised widths, and L2-normalised: `bge-m3` (1024), `qwen3-embed-8b` (4096), `gemini-embed-001` (3072). No hosted *base* (non-instruct) generator was available on the audited OpenRouter catalogue (ADR-0006). Later stages therefore cannot claim a base-versus-instruct contrast on the hosted stack. A later local-base probe at a different W is Stage 3, not a substitute for that missing hosted cell.

4.2 Stage 1: freeze is typical on this instruct model at $W=4096$

Stage 1 generated 48 trajectories on `or-qwen3-8b` at $W=4096$, target $T=131072$ tokens (32 turnovers). The run STATUS is FAILED after a post-generation ledger flush; the trajectory files used downstream are complete. Degeneracy: `s1-degeneracy-20260831T070038Z-b2f6f0c9`. Embeddings: `s1-embed-pilot-core-20260831T064430Z-f05e6a68`.

Does seed identity survive the context horizon? — bge-m3

W = 4,096 tokens. Post-horizon mean gap 0.1569, trend -0.00079 per turnover. Separated at the last observed band.

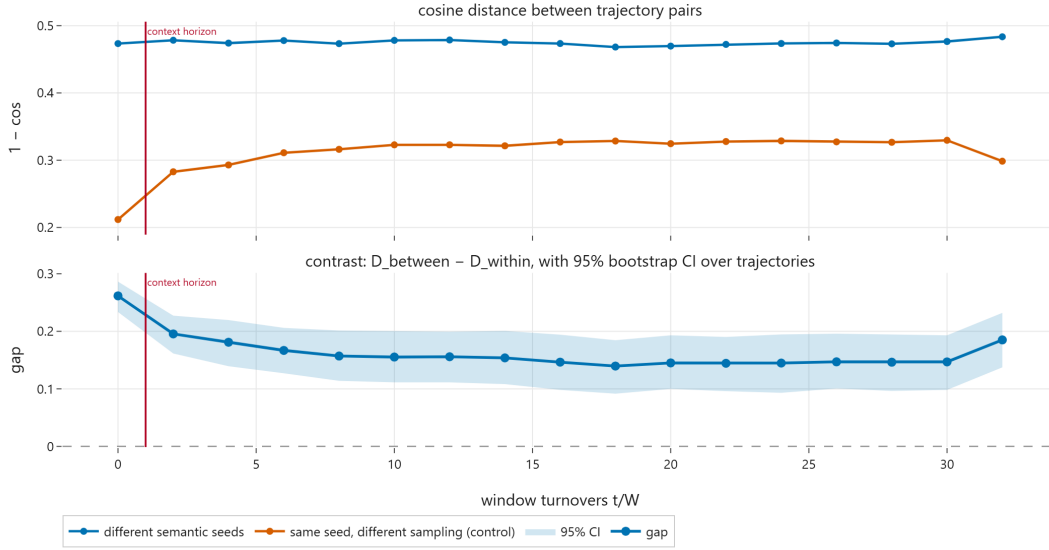


Figure 1: Stage 1 domain-gap versus turnover in **bge-m3** on the 48-trajectory **or-qwen3-8b** pilot. The curve is compatible with seed-family identity surviving the horizon on this degenerate-majority panel. It is not a claim that the same gap exists in a non-looping subsample, at another generator, or under a sliding KV cache. Data: `artifacts/stage-1/separation-bge-m3/seed_separation.data.parquet`. Generate: `s1-pilot-core-20260830T134147Z-f1cce9ec`. Embed: `s1-embed-pilot-core-20260831T064430Z-f05e6a68`. Separation: `s1-separation-bge-m3-20260831T074042Z-9bf54fbf`.

Criterion E9 (textual fixed point): 45/48 trajectories (94%), including 24/24 at $T=0.3$ and 21/24 at $T=1.0$. Criterion E1 (seed-family separation after the horizon): all 17 post-horizon bands have a 95% CI on the domain gap that excludes 0 in both **bge-m3** and **qwen3-embed-8b**. In **bge-m3** the band-0 gap is 0.2616 and the post-horizon plateau is about 0.147.

Figure 1 shows the Stage 1 separation curve in **bge-m3**. It does not show a non-degenerate subsample. MSD scaling exponents $\alpha = 0.244$ (**bge-m3**) and 0.350 (**qwen3-embed-8b**) were computed on this degenerate-majority panel and are not interpreted as confined semantic motion.

A cheap replication arm was planned and never executed. Stage 1 is **PARTIAL** on that criterion.

4.3 Stage 2: the lock is not universal across hosted generators

Table 1 reports textual-fixed-point rates on a four-model axis at $T=0.3$, $W=4096$. **or-gemma-4-31b** is 0/8 (CI [0, 0]). **or-gpt-oss-120b** is 8/8 (CI [1, 1]), under an almost- T caveat documented in the Stage 2 report. **or-muse-glimmer-30b** is 2/8 (CI [0, 0.625]); the rates file counts incomplete trajectories, and only the two long physics cells reached target T . **or-qwen3-8b** on this arm is 7/8 (CI [0.625, 1]). **or-gpt-oss-20b** contributed 0/8 trajectories that reached target T (empty-EOS deaths); the rates CSV’s 2/7 is not used. Across the model-axis file, 11/40 trajectories reached target T .

A prefill-versus-raw mechanism arm on **or-qwen3-8b** did not detect a prefill-driven change in the lock rate: 7/8 versus 8/8, difference CI $[-0.375, 0]$. Provider determinism, measured as

Table 1: Textual fixed-point rates, Stage 2 model axis, $T=0.3$, $W=4096$. Source: `artifacts/stage-2/model-axis/rates/fixed_point_rates.csv`; `s2-rates-20260901T125506Z-41d2e88e`. Gemma’s 0/8 is the existence proof that the Stage 1 freeze is not a property of instruct models as a class.

Generator	n in rates file	fixed-point rate	95% CI
or-qwen3-8b	8	7/8	[0.625, 1]
or-gemma-4-31b	8	0/8	[0, 0]
or-gpt-oss-120b	8	8/8	[1, 1]
or-muse-glimmer-30b	8	2/8	[0, 0.625]
or-gpt-oss-20b	0 reached T	—	—

exact text match on a short probe, was 100% (or-muse-glimmer-30b), 60% (or-qwen3-8b), 20% (or-gemma-4-31b), and 20% (or-gpt-oss-120b). The occupancy stages therefore cannot treat replica pairs as independent draws from a deterministic decoder.

The Stage 2 rate figure exists only as SVG (`artifacts/stage-2/model-axis/rates/fixed_point_rate.svg`). This manuscript uses the CSV table rather than a rasterised substitute.

4.4 Stage 3: no validated MSM; n_{macro} is not an order parameter

Stage 3 applied VAMP/PCCA+ on embeddings of the Stage 1 panel and a local-base probe. The validation table is uniformly negative: `validated=0` in every cell of `artifacts/stage-3/dynamics/k_stability`. Implied n_{macro} is unstable across K and across embedding spaces. Microstate Chapman–Kolmogorov tests fail a 0.15 bar. Hypothesis H1 (metastable semantic states as an MSM object) is unsupported. The Frobenius norm of a probability current computed on *microstates* is not evidence for H4 (non-reversible currents among macrostates).

A local `gemma-3-1b-pt` probe at $W=256$ produced 8/8 degenerate trajectories and 0/8 reviewer-register rows. That cell is a different window from the occupancy panel. It does not fill the missing hosted-base cell from Stage 0, and it does not license a base-versus-instruct claim at $W=4096$.

4.5 Stage 4: lock until $T=1.5$; H5 is absent

Stage 4 crossed temperature and window on `or-qwen3-8b` with $n=4$ trajectories per cell. Table 2 summarises looping rates. At $T \leq 1.0$, both $W=4096$ and $W=8192$ are 4/4 looping (CI [1, 1]). At $T=1.5$, $W=4096$ is 0/4 looping (CI [0, 0]); $W=8192$ is 2/4 (CI [0, 1]).

Clean MSD α is defined only in cells that are not looping. The defined cells are $T=1.5$: at $W=4096$, $\alpha = 0.149$ [0.140, 0.158] in `bge-m3` and $\alpha = 0.277$ [0.241, 0.318] in `qwen3-embed-8b`; at $W=8192$ ($n_{\text{clean}}=2$), $\alpha = 0.146$ [0.086, 0.207] in `bge-m3` and $\alpha = 0.247$ [0.172, 0.322] in `qwen3-embed-8b`. All four estimates are subdiffusive. There is no low- T clean α , so H5 (non-looping confined motion in a temperature window) is absent, not borderline. Figure 2 and Figure 3 are the committed Stage 4 figures. $n=4$ is the whole cell, not a large-sample rate.

4.6 Stage 5: last-band occupancy on the locked panel

Given the lock, Stage 5 asks two occupancy questions on twenty-eight `or-qwen3-8b` trajectories at $W=4096$, $T=0.3$, twelve turnovers. Generate `s5-lock-occupancy-20260905T164327Z-6780902f`; embed `s5-embed-lock-occupancy-20260906T030125Z-eab6e484`; degeneracy `s5-degeneracy-20260906T030`

Table 2: Looping rate by temperature and window, Stage 4, **or-qwen3-8b**, $n=4$ per cell. Source: `artifacts/stage-4/grid/looping_rate_by_cell.csv`. Degeneracy thresholds remain Jac-card 0.083 / late 0.0122.

T	$W=4096$ looping	CI	$W=8192$ looping	CI
≤ 1.0	4/4	[1, 1]	4/4	[1, 1]
1.5	0/4	[0, 0]	2/4	[0, 1]

Looping / fixed-point rate on **or-qwen3-8b** (P1 raw), $n = 4$ per cell

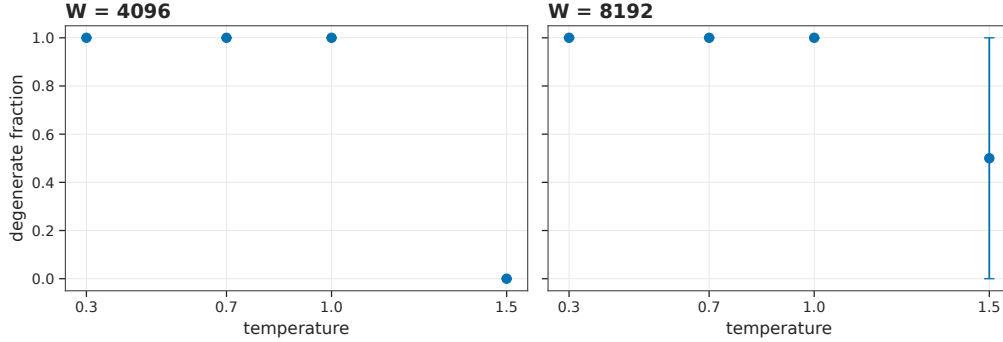


Figure 2: Looping rate versus temperature at two windows. Data: `artifacts/stage-4/grid/looping_rate_vs_T.data.parquet`. Generate: `s4-w4096-new-temps-20260904T103121Z-589c8eb1, s4-w8192-20260904T120057Z-ce82ce55`. The $T=1.5$, $W=4096$ cell is the existence of a non-looping regime on this model, not a mixing-time estimate.

Ten domain seeds (including love) contribute two replicas each; four twin seeds contribute the F6 pairs.

Lock occupancy: 19/20 domain trajectories and 10/10 domain seeds meet the textual-fixed-point rule; love is 1/2.

F4 in two spaces, last band, ten domain seeds, $n=20$ trajectories, $n_{\text{within}}=9$ pairs after physics seed 1 drops from the last-band diagonal: **bge-m3** gap 0.201, 95% CI [0.065, 0.332], separated; **qwen3-embed-8b** gap 0.390, CI [0.151, 0.593], separated. That last-band gap is ensemble distinguishability, not recovered prompt memory (H2). Figure 4 shows the gap versus turnover in those two spaces.

F6 (replica Δ): every last-band CI includes 0 on this panel. Operational collapsed is that CI statement. It is not occupancy of one lock. The Waterloo pair in **bge-m3** has a last-band point estimate near 0.049 whose CI still includes 0. The Stage 5 twin- Δ figure is SVG-only; the three-space figure in Section 4.7 supersedes it for the manuscript.

4.7 Stage 6: occupancy signs in a third embedding space

Stage 6 re-embedded the same 28 trajectories in **gemini-embed-001** and reassembled F4/F6 in three spaces. No new tokens were generated. Hosted Stage 6 spend is \$0. Gemini embedding runs: `s6-embed-third-space-20260906T082301Z-588eff8f`, `s6-embed-third-space-20260906T082628Z-9077d5`

Table 3 reports last-band F4. All three spaces are **separated** under the pre-registered CI rule. The gemini interval [0.029, 0.266] excludes 0 by a whisker. That is an NHST pass, not thick robustness, and not architecture-independence. The three spaces disagree on *level*: 0.201 / 0.390 / 0.150. Sign agreement is the claim. Physics seed 1 has 47 versus 48 chunks, so its last-band

MSD α on non-degenerate trajectories only; $\times$ = $n_{\text{clean}} < 2$ (undefined)

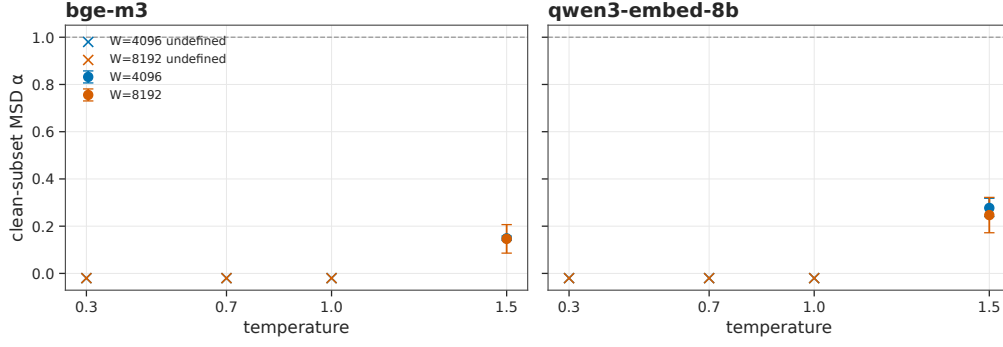


Figure 3: Clean MSD α where defined (non-looping cells only). Data: `artifacts/stage-4/grid/clean_alpha_vs_T.data.parquet`. Missing low- T points are the H5 absence, not a plotting omission.

Ten-domain seed gap vs turnover on or-qwen3-8b (P1 raw), $W=4096$ $T=0.3$

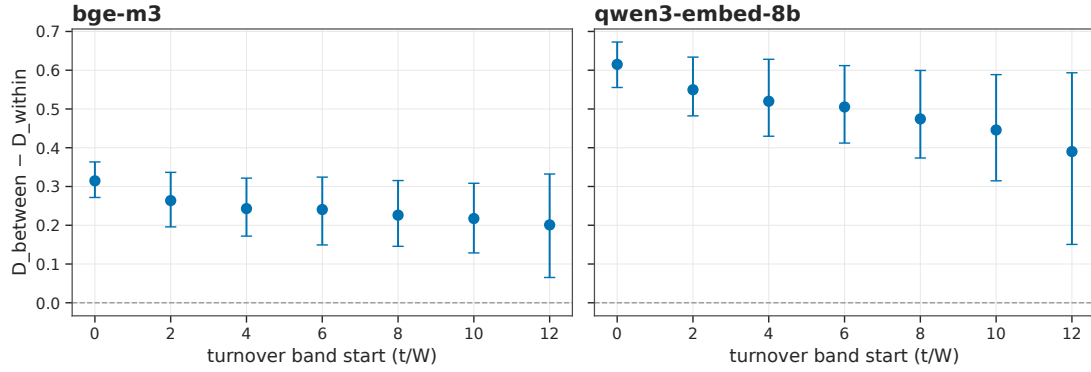


Figure 4: Domain gap versus turnover in **bge-m3** and **qwen3-embed-8b** on the Stage 5 locked panel. Last-band CIs exclude 0 from above in both spaces. The figure does not include **gemini-embed-001** (Stage 6) and does not show a non-looping subsample. Data: `artifacts/stage-5/occupancy/domain_separation_vs_turnover.data.parquet`. Generate: `s5-lock-occupancy-20260905T164327Z-6780902f`.

D_{within} diagonal is undefined (NaN) in every space; F4 uses the remaining $n_{\text{within}}=9$ pairs.

Table 4 reports last-band F6. Gemini all-pairs Δ is -0.012 , CI $[-0.286, 0.223]$ (includes 0). The reactor pair never excluded 0 in any space (0.008, CI $[-0.071, 0.087]$ in gemini). Waterloo gemini last-band Δ is -0.031 , CI $[-0.286, 0.223]$; the same pair at band 0 is 0.033, CI $[0.001, 0.065]$, so the sign flips along the trajectory. Operational collapsed remains “CI includes 0”, not “the two replicas occupy one lock”.

Figure 5 and Figure 6 are the Stage 6 three-space figures. Figure 7 is a last-band cosine-distance matrix in gemini space. It is an illustration of pairwise distances, not a cluster count, and not a 2-D projection used as evidence.

A protocol diagnostic on the same panel, not a headline result: domain trajectories spend a mean fill fraction 0.853, 95% CI $[0.756, 0.936]$, in the last quarter, with mean stop-token fraction 0.359. P1 fill is a confound to name, not a sliding-window measurement.

Table 3: F4 last-band domain gap, ten domain seeds, $n=20$, $n_{\text{within}}=9$. Source: `artifacts/stage-6/occupancy/domain_separation_last_band.csv`. Generate: `s5-lock-occupancy-20260905T164327Z-6780902f`. Gemini embeds: `s6-embed-third-space-20260906T082301Z-588eff8f`, `s6-embed-third-space-20260906T082628Z-9077d587`. Separated means the 95% CI excludes 0 from above.

Space	gap	95% CI	separated
<code>bge-m3</code>	0.201	[0.065, 0.332]	yes
<code>qwen3-embed-8b</code>	0.390	[0.151, 0.593]	yes
<code>gemini-embed-001</code>	0.150	[0.029, 0.266]	yes

Table 4: F6 last-band replica Δ , operational collapsed = CI includes 0. Source: `artifacts/stage-6/occupancy/twin_last_band.csv`. This table is not occupancy of one lock.

Slice	Δ	95% CI	CI includes 0
gemini, all pairs	-0.012	[-0.286, 0.223]	yes
reactor, gemini	0.008	[-0.071, 0.087]	yes
waterloo, gemini last band	-0.031	[-0.286, 0.223]	yes
waterloo, gemini band 0	0.033	[0.001, 0.065]	no

5 Limitations

P1 versus sliding attention. Every afterlife token in this paper is produced by re-prompting the last W tokens. A native sliding KV cache was not measured. ADR-0017 parks that comparison. No sentence here is evidence about sliding attention.

Representation. Occupancy signs agree in `bge-m3`, `qwen3-embed-8b`, and `gemini-embed-001`. Absolute gaps do not: 0.201, 0.390, 0.150. Gemini F4 is an NHST whisker (0.029 on the lower CI). That is not architecture-independence, and Stage 6 closed on that wording. We do not report ARI/NMI across spaces; the cross-embedding check is F4 sign agreement.

Provider non-determinism. Stage 2 exact-match rates were 100/60/20/20 percent across four hosted generators. Replica pairs on `or-qwen3-8b` are not guaranteed independent deterministic draws.

Finite horizon. Occupancy uses twelve turnovers, not an infinite afterlife. Stage 1 reached 32 turnovers on a different seed panel; that length is not reused as an occupancy mixing time.

Metastability versus lock. A textual fixed point is a lock in token space. It is not a validated MSM macrostate. Stage 3 found `validated=0` in every cell. This paper does not use “metastable semantic states” as an established finding.

One generator, one protocol, one temperature for occupancy. F4 and F6 are `or-qwen3-8b`, P1, `raw_completion`, served provider Alibaba, $T=0.3$, $W=4096$. P1 is the re-prompt axis; `raw_completion` is the continuation mechanism. They are not synonyms. Stage 2 already measured prefill versus raw on this generator (7/8 versus 8/8, difference CI [-0.375, 0]); that

Ten-domain last-band gap in three embedding spaces

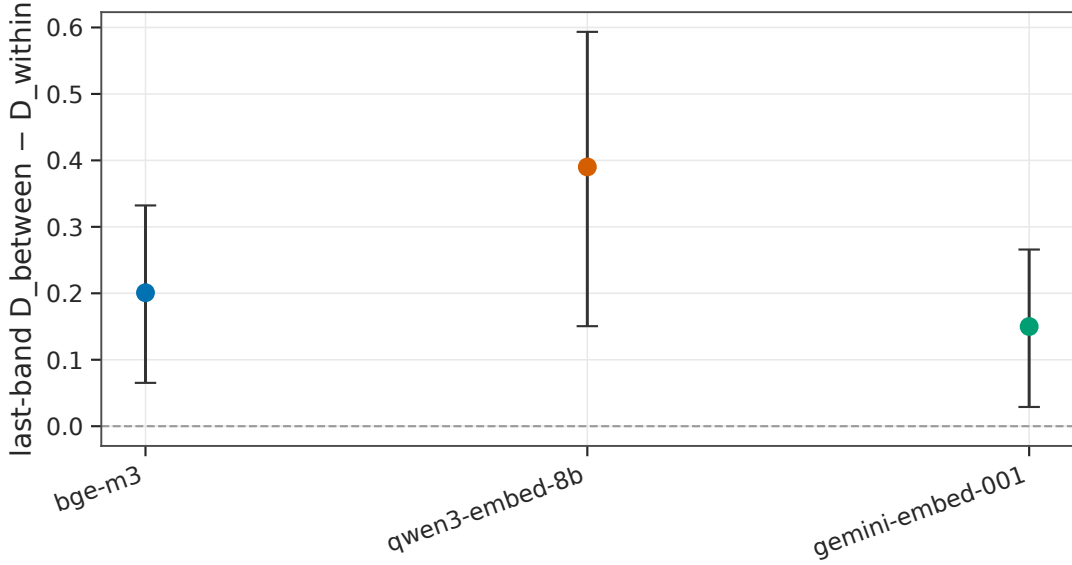


Figure 5: F4 domain gap versus turnover in three embedding spaces on the same 28 or-qwen3-8b trajectories. Last-band CIs exclude 0 in all three spaces; gemini is the whisker. Levels 0.201 / 0.390 / 0.150 are not interchangeable. Data: artifacts/stage-6/occupancy/domain_gap_three_spaces.data.parquet.

arm is not the occupancy protocol. Stage 2 already shows a generator where the lock rate is 0/8. Stage 4 already shows $T=1.5$, $W=4096$ with 0/4 looping. Those cells are not occupancy measurements.

Replica $n=2$. Each domain seed has two trajectories. F6 CIs are wide. A CI that includes 0 can be an underpowered non-detection.

F4 whisker and physics seed 1. Gemini last-band F4 lower CI is 0.029. Physics seed 1 last-band D_{within} is NaN because of a 47 versus 48 chunk mismatch.

F6 is not one lock. Operational collapsed means the last-band Δ CI includes 0. Waterloo in gemini flips sign from band 0 (0.033, CI excludes 0) to the last band (-0.031 , CI includes 0).

Degeneracy is the sample. 19/20 domain trajectories are textual fixed points. Occupancy is a fact about locked text, not about a clean subsample that was held out. Thresholds 0.083 / 0.0122 were not moved; love s1 is kept.

No hosted base model. Stage 0 found none. The local-base probe is `gemma-3-1b-pt` at $W=256$, not the occupancy window.

6 Discussion

What is established, on this generator and protocol: after the seed leaves W , the typical $T=0.3$ trajectory is a textual lock; last-band centroids still show an ensemble domain gap across ten

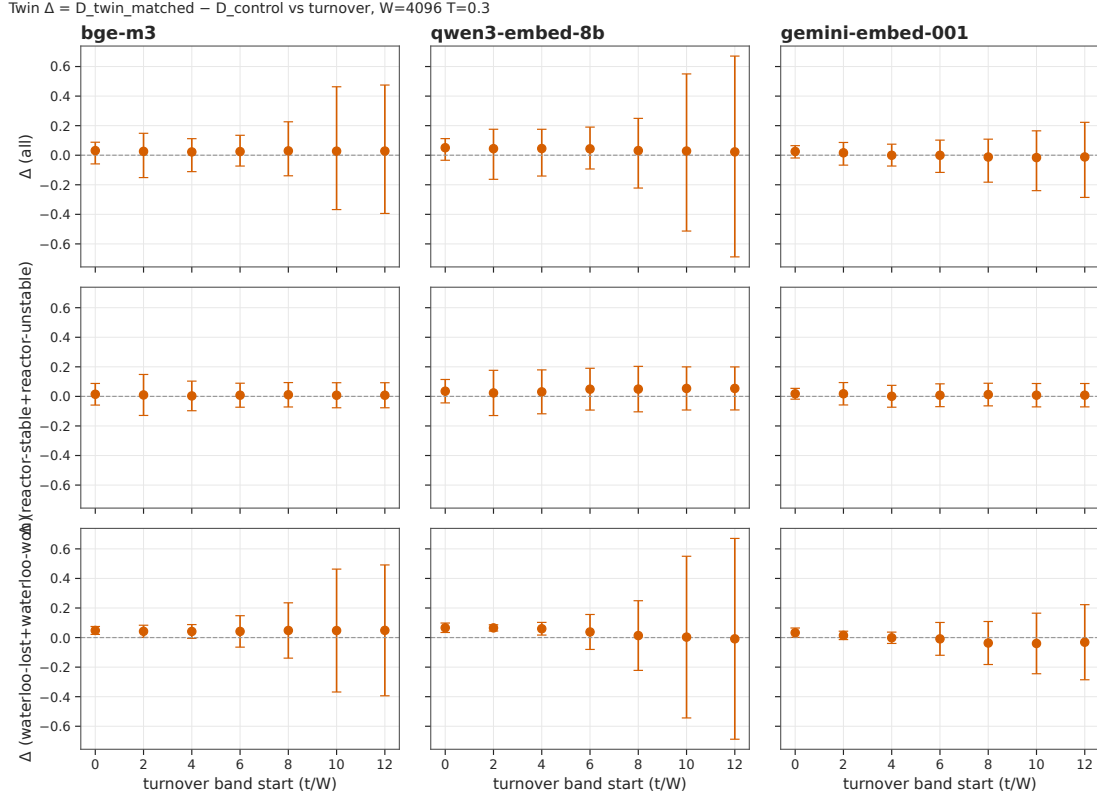


Figure 6: F6 replica Δ versus turnover in three spaces. Last-band CIs include 0. Waterloo in gemini flips sign from band 0 to the last band. Operational collapsed is that CI statement, not a shared lock. Data: `artifacts/stage-6/occupancy/twin_delta_vs_turnover.data.parquet`.

seed domains in three embedding spaces (not recovered prompt memory, H2); replica Δ CIs at the last band include 0 and do not justify a one-lock headline.

What is not established: H1 (validated MSM macrostates); H5 (a temperature window of non-looping confined motion); we cannot claim architecture-independence of occupancy; a result for language models as a class; P1 versus sliding attention; mixing times; non-reversible macrostate currents.

Zekri et al. [2024] remain the right uniqueness null. Domain-tagged last-band gaps on a locked panel are compatible with more than one long-run occupancy region in embedding space. That compatibility is not a proof that the token-level chain has several stationary measures. It is a measurement of where locked trajectories sit after twelve turnovers.

The honest next measurement, if any, is not another embedding of the same 28 rows. It is a second generator, a sliding-cache contrast, or a non-looping occupancy panel at $T=1.5$. Those are parked, not silently folded into this manuscript.

7 Reproducibility

Experiments through Stage 6 closed on git commit `ad1f244`. A run is `configs/**` plus that SHA plus a seed. Raw trajectories live under `runs/` (gitignored). Publication figures and tidy frames live under `artifacts/`. The command `afterlife reproduce <run_id>` is the intended re-derivation path for a generation or analysis run.

Stage 7 adds this manuscript. It does not add a `configs/stages/stage7_*.yaml` and does not mint a dummy run to satisfy a generation gate. Hosted Stage 7 spend is \$0.

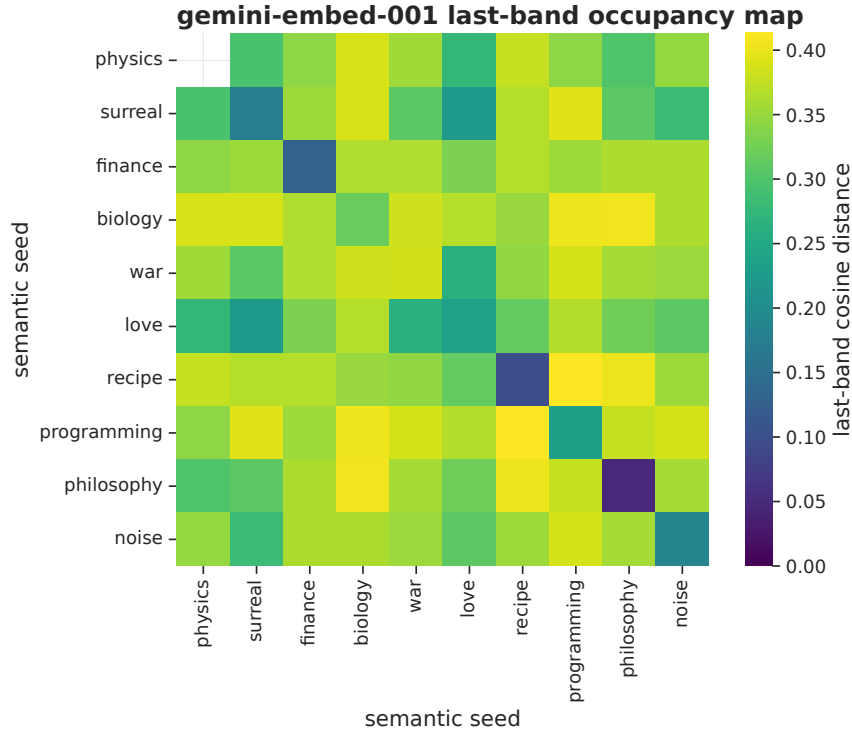


Figure 7: Last-band cosine distances in **gemini-embed-001** (illustration of the pairwise matrix, not a cluster count, not a 2-D projection). Physics seed 1 is the incomplete last band. Data: artifacts/stage-6/occupancy/last_band_distance_matrix_gemini_embed_001.data.parquet.

Primary **run_ids** cited in the text:

- Stage 1 generate: **s1-pilot-core-20260830T134147Z-f1cce9ec**
- Stage 1 degeneracy: **s1-degeneracy-20260831T070038Z-b2f6f0c9**
- Stage 1 embed: **s1-embed-pilot-core-20260831T064430Z-f05e6a68**
- Stage 1 separation: **s1-separation-bge-m3-20260831T074042Z-9bf54fbf**
- Stage 1 geometry: **s1-geometry-bge-m3-20260831T074420Z-d3c0204a**
- Stage 2 model axis: **s2-model-axis-20260901T015457Z-ab59afc8**
- Stage 2 mechanism: **s2-mechanism-20260901T071519Z-dfbb173a**
- Stage 2 determinism: **s2-audit-determinism-20260901T015221Z-ab59afc8**
- Stage 2 rates: **s2-rates-20260901T125506Z-41d2e88e**
- Stage 3 local base: **s3-local-base-20260901T184812Z-cc80633b**
- Stage 3 dynamics: **s3-dynamics-20260901T204521Z-f21d5908**, **s3-dynamics-20260901T204549Z-b7c40**
- Stage 4: **s4-w4096-new-temps-20260904T103121Z-589c8eb1**, **s4-w8192-20260904T120057Z-ce82ce55**
- Stage 5 generate: **s5-lock-occupancy-20260905T164327Z-6780902f**
- Stage 5 embed: **s5-embed-lock-occupancy-20260906T030125Z-eab6e484**

- Stage 5 degeneracy: `s5-degeneracy-20260906T030145Z-deb4c3bd`
- Stage 6 gemini: `s6-embed-third-space-20260906T082301Z-588eff8f`, `s6-embed-third-space-202609`

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